



## OPEN MPPT algorithms for grid-connected solar systems including deep learning approaches

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Photovoltaic (PV) systems, which are the most abundant renewable resources, convert solar radiation into electricity through solar cells but cannot consistently operate at the Maximum Power Point (MPP). Therefore, an external controller using Maximum Power Point Tracking (MPPT) is required. The accuracy and efficiency of this control directly influence system performance, and optimised algorithms can significantly improve results. This study presents a comparative analysis of MPPT algorithms based on efficiency, total harmonic distortion (THD), oscillation behaviour, computational complexity, relative power loss, and relative power gain. The MPPT methods include conventional Perturb and Observe (P&O) and Incremental Conductance (INC); meta-heuristic techniques such as Grey Wolf Optimisation (GWO), Fuzzy Logic (FL), and Particle Swarm Optimisation (PSO); and learning based approaches including Artificial Neural Network (ANN), Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), and Adaptive Neuro-Fuzzy Inference System (ANFIS). Results reveal that GWO, PSO, and learning-based approaches offer the highest performance, offering around 99% efficiency, low oscillations, favourable THD, and rapid decision-making. While P&O and INC reach nearly 98.5% efficiency, their effectiveness is limited by stronger oscillatory behaviour. FL causes the highest THD, and its high computational complexity and reduced efficiency limit suitability under rapidly changing operating conditions.

**Keywords** Maximum power point tracking, Photovoltaic panels, Algorithms, Deep learning

### Abbreviations

|                 |                                       |
|-----------------|---------------------------------------|
| PV              | Photovoltaic                          |
| PSC             | Partial Shading Condition             |
| RES             | Renewable Energy System               |
| SPP             | Solar Power Plant                     |
| I-V             | Current and Voltage                   |
| P-V             | Power and Voltage                     |
| CO <sub>2</sub> | Carbon Dioxide                        |
| MPP             | Maximum Power Point                   |
| GMPP            | Global Maximum Power Point            |
| MPPT            | Maximum Power Point Tracking          |
| THD             | Total Harmonic Distortion             |
| P&O             | Perturb and Observe                   |
| INC             | Incremental Conductance               |
| GWO             | Grey Wolf Optimisation                |
| FL              | Fuzzy Logic                           |
| PSO             | Particle Swarm Optimisation           |
| ANN             | Artificial Neural Network             |
| RNN             | Recurrent Neural Network              |
| LSTM            | Long Short-Term Memory                |
| BiLSTM          | Bidirectional LSTM                    |
| ANFIS           | Adaptive Neuro-Fuzzy Inference System |
| CV              | Constant Voltage                      |
| FOCV            | Fractional Open Circuit Voltage       |

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|                   |                                     |
|-------------------|-------------------------------------|
| FSCC              | Fractional Short Circuit Current    |
| WOA               | Whale Optimisation Algorithm        |
| PSOMA             | PSO Memetic Algorithm               |
| GEP               | Gene Expression Programming         |
| MSOA              | Modified Seagull Optimisation       |
| EWOA              | Enhanced Whale Optimisation         |
| ESM               | Enhanced Slime Mold Optimisation    |
| ABC               | Ant Bee Colony Optimisation         |
| ABSO              | Artificial Bee Swarm Optimisation   |
| GA                | Genetic Algorithm                   |
| CM                | Covariant Matrix                    |
| FOPI              | Fractional Order PI                 |
| GOA               | Grasshopper Optimisation Algorithm  |
| GWEBO             | Grey Wolf Election Optimisation     |
| APSO              | Advanced PSO                        |
| ASO               | Atom Search Optimisation            |
| BAT               | Bat Algorithm                       |
| EPSO              | Enhanced PSO                        |
| IPSO              | Improved PSO                        |
| DNN               | Deep Neural Network                 |
| DRL               | Deep Reinforcement Learning         |
| AVOA-RNN          | African Vulture Optimisation – RNN  |
| RCC               | Ripple Correlation Control          |
| DA                | Dragonfly Algorithm                 |
| MFOA              | Moth–Flame Optimization Algorithm   |
| SSOA              | Salp Swarm Optimization Algorithm   |
| ML                | Machine Learning                    |
| $V_{OC}$          | Open Circuit Voltage                |
| $I_{SC}$          | Short Circuit Current               |
| $V_{MPP}$         | Voltage at MPP                      |
| $I_{MPP}$         | Current at MPP                      |
| $V_{pv}$          | Panel Voltage                       |
| $I_{pv}$          | Panel Current                       |
| $P_{pv}$          | Panel Power                         |
| $P_{pv,ideal}$    | Ideal Panel Power                   |
| $P_{pv,real}$     | Extracted, Real Output Power        |
| $P_{pv,proposed}$ | Proposed Algorithm Output Power     |
| $P_{pv,base}$     | Base Output Power                   |
| $I_s$             | Output Current                      |
| $I_{ph}$          | Photocurrent                        |
| $I_D$             | Diode Current                       |
| $I_p$             | Parallel Current                    |
| $I_{SD}$          | Reverse Saturation Current          |
| $R_{sn}$          | Snubber Resistance                  |
| $R_{on}$          | Opening Resistance                  |
| $R_s$             | Series Resistance                   |
| $R_p$             | Parallel Resistance                 |
| $q$               | Electron Charge                     |
| $n$               | Ideality Factor                     |
| $k$               | Boltzmann's Constant                |
| $T$               | Temperature                         |
| $D$               | Duty Cycle                          |
| $V_o$             | Output Voltage                      |
| $L$               | Inductance                          |
| $f_s$             | Switching Frequency                 |
| $\Delta I_L$      | Output Ripple Current               |
| $C_0$             | Output Capacitance                  |
| $I_0$             | Output Current                      |
| $\Delta V_0$      | Output Ripple Voltage               |
| RMS               | Root Mean Square                    |
| $I_n$             | RMS Value of Harmonic               |
| $I_F$             | RMS Value of Fundamental Current    |
| $T_s$             | Sampling Time                       |
| IGBT              | Insulated - Gate Bipolar Transistor |
| PI                | Proportional, Integral              |
| $K_p$             | Proportional Constant               |
| $K_i$             | Integral Constant                   |
| Trimf             | Triangular Membership Function      |

In recent years, the substantial rise in global energy demand has been primarily driven by the continuous growth of the world population. However, using fossil fuels to obtain energy increases the emission of CO<sub>2</sub> and greenhouse gases. CO<sub>2</sub> and greenhouse gases cause global warming and climate change<sup>1,2</sup>. The Paris Agreement was announced in 2015 to combat climate change. This global agreement accelerated the usage of renewable energy systems such as solar, wind, and geothermal energy<sup>3</sup>.

SPPs, one of the main RESs, convert solar irradiance to electrical energy via solar cells. MPPT represents a critical control strategy in PV energy systems, aimed at continuously harvesting the maximum attainable power under varying environmental conditions. This is achieved by precisely modulating the PV array's operating parameters—typically voltage or current—to ensure alignment with the system's MPP on the P–V characteristic curve. By effectively matching the PV generator's I–V operating point to the load profile, MPPT algorithms significantly enhance overall energy conversion efficiency<sup>4</sup>.

In literature, MPPT algorithms are categorised into three categories: Conventional algorithms, meta-heuristic algorithms, and learning based algorithms. Conventional algorithms are traditional algorithms that are FOCV<sup>5</sup>, FSCC<sup>6</sup>, P&O<sup>7</sup>, and INC<sup>8</sup>. However, these algorithms produce oscillations around MPP. Moreover, meta-heuristic and intelligent algorithms include GWO<sup>9</sup>, FL type-1<sup>10</sup>, type-2<sup>11</sup>, type-3<sup>12</sup>, Hybrid FL Algorithm<sup>13</sup>, PSO<sup>14</sup>, Flower Pollination<sup>15</sup>, GA<sup>16</sup>, Cuckoo Search Algorithm<sup>17</sup>, Arithmetic Optimisation<sup>18</sup>, Adaptive Snake Algorithm with P&O<sup>19</sup>, Hill Climbing Algorithm<sup>20</sup>, GWO with WOA<sup>21</sup>. Learning based algorithms use ANN<sup>22,23</sup>, a type of RNN: LSTM<sup>24</sup>, ANFIS<sup>25</sup>, and GEP with ANFIS<sup>26</sup>. In Table 1, MPPT algorithms in the literature have been examined, and the innovations they brought to the literature have been explained.

In this paper, a comprehensive comparative evaluation of widely used MPPT algorithms for grid-connected PV systems is conducted. Conventional (P&O, INC), metaheuristic (GWO, PSO), intelligent (FL) and learning based (, ANN, LSTM, BiLSTM, and ANFIS) approaches are systematically examined under different operating conditions, including sinusoidal irradiation and PSC scenarios. The algorithms are evaluated in terms of THD, and computational complexity. In addition, practical implementation aspects such as typical hardware requirements, inference latency, memory usage, and real-time deployability are explicitly considered to assess the feasibility of deploying each algorithm on embedded control platforms. Although THD is primarily associated with the output quality of the inverter, it is indirectly influenced by the dynamic behavior of MPPT algorithms. MPPT techniques that induce rapid or unstable voltage and current variations can introduce fluctuations at the inverter input, which may propagate to the output stage and increase the harmonic content. For this reason, THD is incorporated in this study as a secondary yet relevant performance metric to capture the impact of MPPT dynamics on overall power quality. By jointly considering algorithmic performance, operating conditions, and hardware constraints, this work provides a holistic assessment framework and practical guidelines for selecting appropriate MPPT algorithms in grid-connected PV applications.

The main contributions and contents of this research can be summarised as follows:

- Theoretical modelling and simulation of a grid-connected PV system are performed.
- The contribution of this paper is the evaluation of a wide range of MPPT controllers, from conventional methods like P&O and INC to meta-heuristic methods such as GWO, FL, and PSO, and learning based approaches including ANN, LSTM, BiLSTM, and ANFIS, aiming to compare their performance under sinus irradiance test, and PSC, thereby supporting the scientific community in enhancing the efficiency and reliability of solar energy conversion systems.
- ANN, LSTM, BiLSTM, and ANFIS models are trained using synthetic irradiance and temperature datasets to evaluate their predictive capabilities in the sinus irradiance test, and PSCs.
- The core contribution of this study is a thorough comparative analysis of the selected MPPT controllers, focusing on critical performance metrics including efficiency, oscillation amplitude, relative power loss, relative power gain, and computational complexity. In addition to primary metrics such as efficiency, oscillation amplitude, relative power loss, relative power gain, and computational complexity, this study also evaluates THD. The study identifies learning-based MPPT algorithms as promising solutions for robust and accurate power point tracking in PV systems.

The paper structure is organised as follows: The Introduction is given in the section “[Introduction](#)”. The general overview of the grid-connected PV system and its parts (PV panel, boost converter, inverter, and MPPT algorithms) are presented in the section “[Methods](#)”. In the section “[MPPT algorithms](#)”, the MPPT algorithms used in this paper are explained. Section “[Results and discussion](#)” presents results and discussion of the performance of MPPT algorithms based on efficiency, oscillation, THD, relative power loss, relative power gain, and computational complexity. The conclusion is given in the section “[Conclusion](#)”.

## Methods

### Theoretical background

The general overview of grid-connected PV systems is shown in Fig. 1. The system has five main parts: PV panels, boost converter, inverter, inductive filter, and grid.

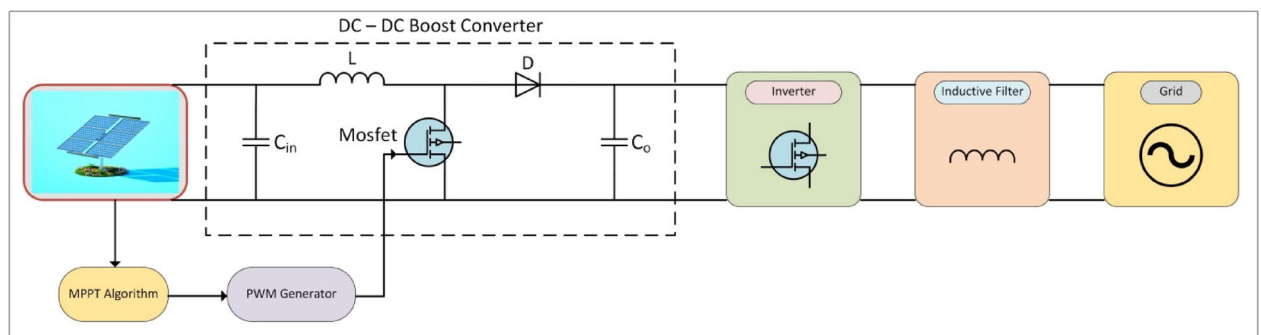
#### *PV panel*

The equivalent circuit of a PV cell is illustrated in Fig. 2. A PV cell consists of semiconductor materials that convert solar energy into electrical energy. When sunlight strikes the surface of a solar cell, it excites electrons, causing electricity to flow through the semiconductor. Solar panels are composed of multiple interconnected solar cells working together to increase voltage or current output. The electrical behaviour of a solar panel is often modelled using the equivalent circuit of a single PV cell. This equivalent circuit typically includes one or more diodes, as well as series and parallel resistances.

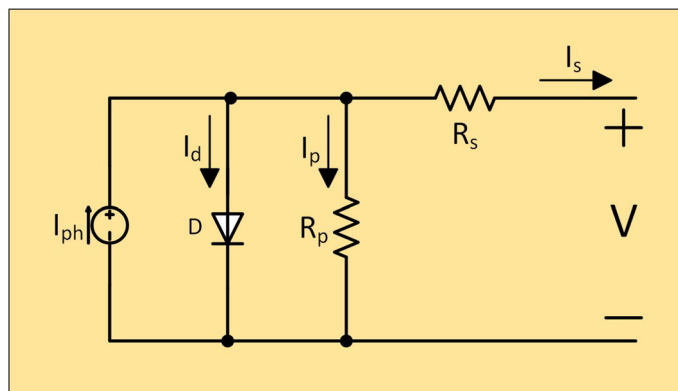
| Conventional Algorithms                                         | Study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P&O <sup>27</sup>                                               | The classical P&O algorithm is optimized with a technique called “Ladder Configuration”. The algorithm performed 99.54% power tracking efficiency in PSCs.                                                                                                                                                                                                                                                                                                                                                                                             |
| P&O <sup>28</sup>                                               | Research shows that the classical P&O algorithm is not enough for MPPT. Therefore, the P&O algorithm is modified by considering additional conditions, such as drift and power loss problems.                                                                                                                                                                                                                                                                                                                                                          |
| Modified INC <sup>29</sup>                                      | The modified INC algorithm is represented, and the performance of the proposed method is compared with that of conventional INC. The proposed algorithm clearly gives high efficiency compared to the classical INC.                                                                                                                                                                                                                                                                                                                                   |
| Meta-Heuristic Algorithms                                       | Study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Hybrid PSO & INC <sup>30</sup>                                  | The hybrid algorithm, which consists of PSO and INC is proposed, and the proposed algorithm performs with high efficiency.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Disturbance Observer – Based Sliding Mode Control <sup>31</sup> | Disturbance Observer – Based Sliding Mode Control is proposed to eliminate the panel output voltage sensor for MPPT. The performance of the proposal is examined both theoretically and experimentally.                                                                                                                                                                                                                                                                                                                                                |
| MSOA <sup>32</sup>                                              | The MSA algorithm is presented to improve the MPPT algorithm efficiency under PSC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| EWOA <sup>33</sup>                                              | EWOA and WOA are compared. The proposed algorithm (EWOA) performed power tracking efficiency is up to 99.93% and the convergence time is 0.22 s.                                                                                                                                                                                                                                                                                                                                                                                                       |
| ESM <sup>34</sup>                                               | The ESM algorithm is proposed and compared with ABC with P&O, ABSO, GA, CM, APSO, and ABC algorithms under PSC. The proposed ESM algorithm outperformed the other algorithms.                                                                                                                                                                                                                                                                                                                                                                          |
| ASO <sup>35</sup>                                               | The ASO algorithm is hybridized with PI, FOPI, and FL under modified MPPT controller conditions. The proposed controllers are evaluated under four different scenarios. Overall, the ASO algorithm combined with the FL approach outperformed the other algorithms.                                                                                                                                                                                                                                                                                    |
| GWEBO <sup>36</sup>                                             | The GWEBO algorithm is compared with meta-heuristic and intelligent methods. The proposed GWEBO algorithm exhibits a lower error rate compared to the other algorithms.                                                                                                                                                                                                                                                                                                                                                                                |
| T-S Fuzzy Observer-Based Algorithm <sup>37</sup>                | T-S Fuzzy Observer-Based Algorithm is a modified version of FL. The proposed algorithm is compared with P&O, PSO, and classical FL. The algorithms are tested under three different PSC. The proposed algorithm has excellent tracking capability in all conditions.                                                                                                                                                                                                                                                                                   |
| BAT <sup>38</sup>                                               | Bat algorithm is hybridised with P&O, INC, and Beta algorithms. The algorithm is tested on different photovoltaic panel models under varying environmental conditions. The advantages and disadvantages of each version of the Bat algorithm are explained.                                                                                                                                                                                                                                                                                            |
| EPSO Algorithm <sup>39</sup>                                    | EPSO is proposed as one of the types of PSO. The proposed algorithm (EPSO) is compared with PSO, GWO, BAT, GOA, and IPSO algorithms.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Poincaré Map Algorithm <sup>40</sup>                            | The Poincaré Map Algorithm is proposed to determine MPP in SPP. The proposed method was compared with some other meta-heuristic algorithms and performed with high efficiency.                                                                                                                                                                                                                                                                                                                                                                         |
| Quantum MPPT Algorithm <sup>41</sup>                            | The quantum algorithm is developed and tested for the MPPT problem using quantum PSO. The proposed method is tested under three different conditions, namely normal, high temperature and PSC.                                                                                                                                                                                                                                                                                                                                                         |
| Walrus Algorithm <sup>42</sup>                                  | The results are compared with those obtained using PSO, GWO, and the Cuckoo Search algorithm. According to the findings, the Walrus algorithm outperformed the other relevant algorithms.                                                                                                                                                                                                                                                                                                                                                              |
| Ali Baba and Forty Thieves Optimisation <sup>43</sup>           | The PID and improved Ali Baba and Forty Thieves Optimisation is used for MPPT in an EV charging station powered by renewable energy. The algorithm is evaluated under four operating conditions: constant temperature (25 °C) with varying irradiance; varying temperature and irradiance; a reference output voltage of 500 V; and night-mode operation with no power generation. A comparative analysis with the Cuckoo Search and GWO algorithms shows that the proposed method consistently outperforms the others under all operating conditions. |
| Hippopotamus Algorithm <sup>44</sup>                            | The proposed Hippopotamus algorithm is compared with GWO, Cuckoo Search, and PSO. The results show that the proposed algorithm achieved high performance, demonstrating an efficiency of 89.92%.                                                                                                                                                                                                                                                                                                                                                       |
| Learning Based Algorithms                                       | Study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| AVOA-RNN <sup>45</sup>                                          | The proposed algorithm is a hybridisation of a metaheuristic algorithm (AVOA) with a deep learning method (RNN). The proposed algorithm gives the best solution with 99.81% tracking capability.                                                                                                                                                                                                                                                                                                                                                       |
| DNN Algorithm <sup>46</sup>                                     | Deep learning control-based MPPT is proposed and compared with P&O algorithm.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| RNN <sup>47</sup>                                               | The ML algorithms, namely Random Forest, Feedforward Neural Network, and RNN are compared for MPPT in a PV panel. The random forest algorithm shows superior performance in terms of execution time.                                                                                                                                                                                                                                                                                                                                                   |
| Supervised Machine Learning <sup>48</sup>                       | The supervised machine learning is proposed in a solar plant for MPPT analysis. The performance efficiency of the system is 99.794%.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Stack LSTM Algorithm <sup>49</sup>                              | The novel deep learning based stack LSTM is developed for the MPPT controller to maximize power in a 100 kW grid-connected PV system. The performance of the developed method is compared with the Feed Forward-Deep Neural Network and P&O.                                                                                                                                                                                                                                                                                                           |
| LSTM Algorithm <sup>50</sup>                                    | For the training process, real-world data is used. This data configuration improves the system reliability. Data management capability and power tracking efficiency are compared with P&O and ANN. In both categories, LSTM performs excellently compared to P&O and ANN.                                                                                                                                                                                                                                                                             |
| DRL <sup>51</sup>                                               | This paper proposed a DRL-based approach and compared the DRL algorithm with FOCV, FSOC, P&O, and PSO methods. The DRL-based MPPT model was described in detail and demonstrated superior efficiency compared to the other algorithms.                                                                                                                                                                                                                                                                                                                 |
| Decision Tree Machine Learning Algorithm <sup>52</sup>          | This paper introduces machine learning instead of using deep learning techniques specifically. Also, instead of using the hidden layers technique in deep learning methods, the decision tree method is used to decide the MPP process.                                                                                                                                                                                                                                                                                                                |
| GWO – ANFIS <sup>53</sup>                                       | The novel hybrid GWO-ANFIS algorithm is represented. The algorithm is tested under solar irradiance and temperature values that varied at specific time intervals.                                                                                                                                                                                                                                                                                                                                                                                     |

| Comparative Analysis                                                          | Study                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Comparative Analysis <sup>54</sup>                                            | The conventional algorithms (P&O, INC), meta-heuristic algorithms (PSO, FL), and hybridization of them (P&O – PSO, FL – PSO), and learning based techniques (ANN, ANFIS) with metaheuristic algorithm hybridisation (ANN – PSO) are examined. In all algorithms, the advantages and disadvantages of each are mentioned.                                                                 |
| Comparative Analysis <sup>55</sup>                                            | Various MPPT techniques, such as Beta Method, Hill Climbing, hybridized with P&O, INC, FOCV, FSCC, FL, and neural networks, are investigated in detail.                                                                                                                                                                                                                                  |
| Comparative Analysis <sup>56</sup>                                            | MPPT techniques such as P&O, INC, and CV, are compared. The P&O, INC, and CV methods achieve efficiencies of 97.2%, 98.5%, and 92.7%, respectively.                                                                                                                                                                                                                                      |
| Comparative Analysis <sup>57</sup>                                            | MPPT algorithms are classified into four categories: classical, intelligent, optimisation-based, and hybrid. The classical MPPT algorithms are examined, including P&O, INC, FOCV, FSCC, and RCC; the intelligent methods include FL and ANN; the optimisation-based approaches include PSO, GWO, ACO, and ABC; and the hybrid methods include FSCC–P&O, FL–P&O, PSO–P&O, and Fuzzy PSO. |
| Comparative Analysis of Nature-Inspired MPPT Algorithms <sup>58</sup>         | This study introduces an optimized approach for MPPT along with fault classification in a PV system. The nature-inspired DA, MFOA, GOA, and SSOA are employed to assess the tracking efficiency of solar PV systems. The proposed SSOA has maximum performance compared with other meta-heuristic techniques, with 98.75% efficiency.                                                    |
| Comparative Analysis of Artificial Intelligence MPPT Techniques <sup>59</sup> | Various artificial intelligence-based MPPT methods have been compared with conventional, meta-heuristic, and hybrid approaches. The findings indicate that AI-based MPPT techniques offer superior efficiency compared to other methods; however, their implementation requires more advanced system capabilities.                                                                       |

**Table 1.** MPPT literature Analysis.



**Fig. 1.** PV Panel System.



**Fig. 2.** Equivalent Circuit of a PV Cell.

Equation (1) shows the output current of the PV panel.

$$I_s = I_{ph} - I_{SD} \left[ \exp \left( \frac{q(V_{PV} + I_s R_s)}{nkT} \right) - 1 \right] - \frac{V_{pv} + I_s R_s}{R_p} \quad (1)$$

**Sinus irradiance test** The output power of solar panels depends on solar radiation and panel temperature. In this context, two different solar radiation profiles are generated to test the algorithms. As the first profile, a si-

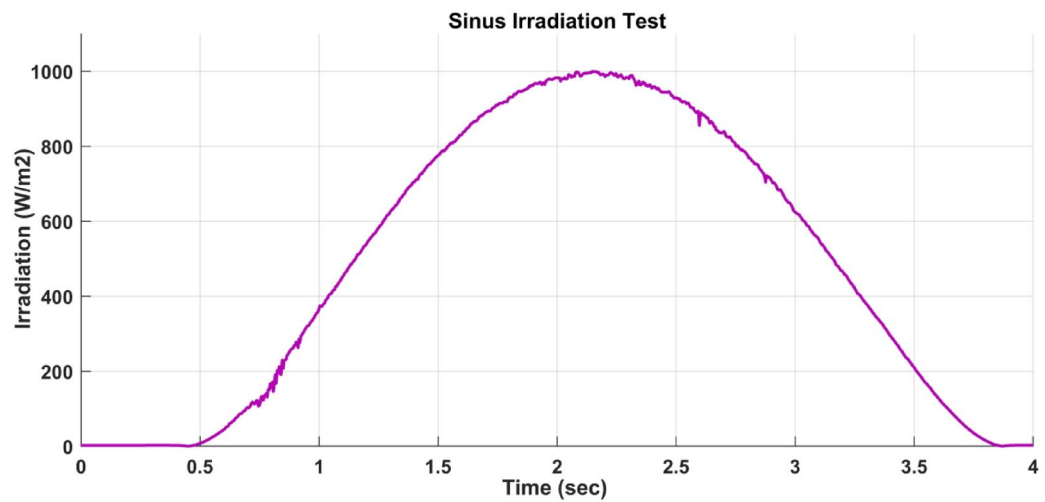


Fig. 3. Sinus Irradiation Test Curve.

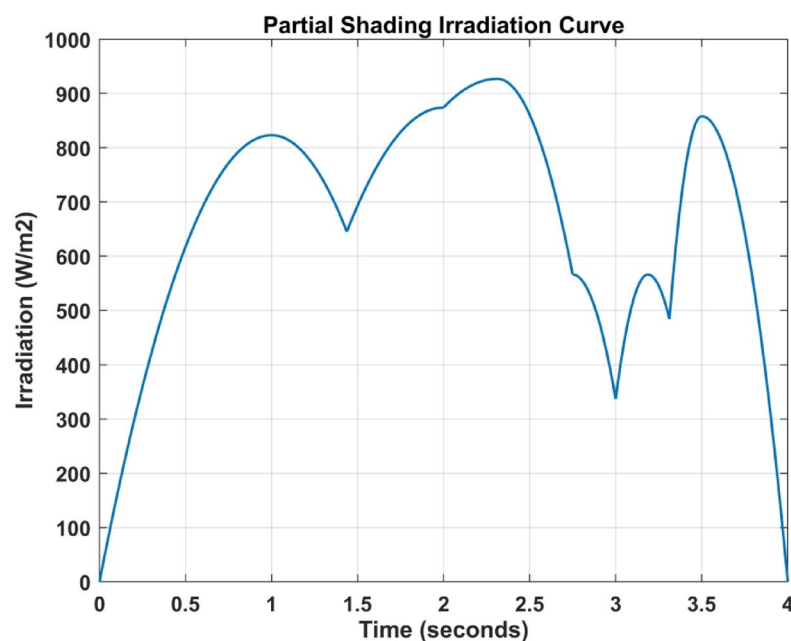
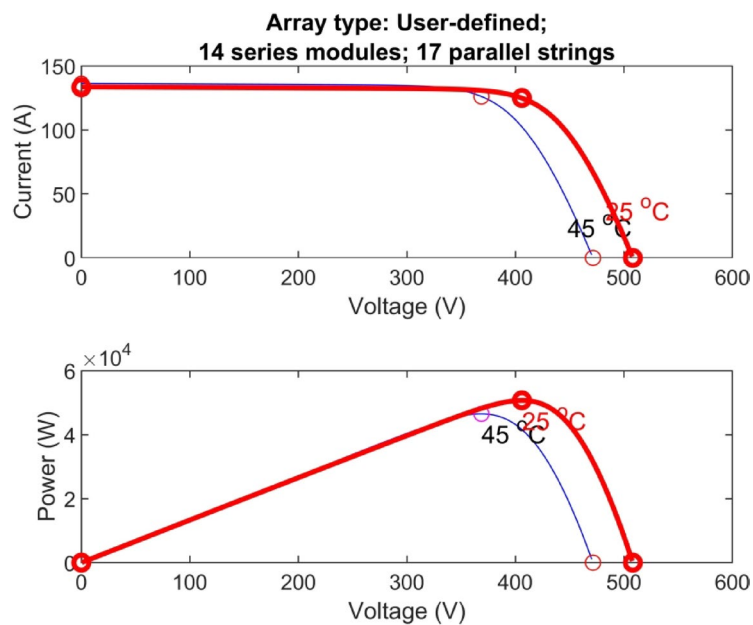


Fig. 4. Typical Solar Irradiance Curve with PSC.

nusoidal solar radiation test is employed. This test represents the typical daily increase in solar radiation under normal environmental conditions. Figure 3 illustrates the sinus irradiation test curve<sup>60–62</sup>.

**Partial shading condition** PSC is one of the most significant factors affecting power efficiency in solar systems. This condition occurs when shadows are cast on solar panels for any reason. Objects such as clouds, trees, buildings, etc., can cast shadows on solar panels, reducing the solar radiation falling on the solar panel and reducing efficiency. Furthermore, sudden PSC can disrupt the ability of the algorithm to extract maximum power, reducing efficiency. Therefore, MPPT algorithms must be prepared for PSCs<sup>63</sup>. In the current study, a 4-second artificial solar irradiation curve is created, and PSC is assumed in the system in all parts. The temperature is constant at 25 °C. The Typical Solar Irradiance Curve with PSC is demonstrated in Fig. 4<sup>64</sup>.

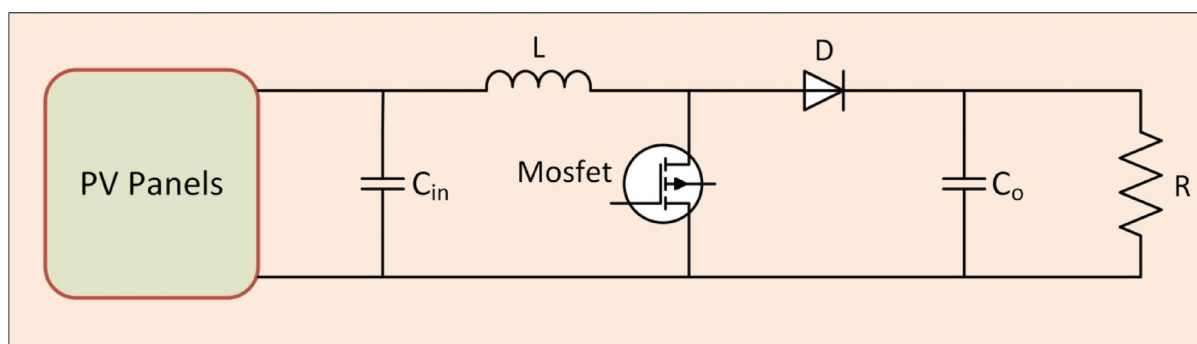
I–V characteristic and P–V curves of the PV panel are illustrated in Fig. 5. The PV panel parameters are provided in Table 2.



**Fig. 5.** PV Panel Voltage–Current and Voltage–Power Graphs.

| Parameters                                                    | Value    |
|---------------------------------------------------------------|----------|
| Maximum Power (W)                                             | 213.15   |
| Number of Cells Per Module                                    | 60       |
| $V_{OC}$ (V)                                                  | 36.3     |
| $I_{SC}$ (A)                                                  | 7.84     |
| $V_{MP}$ (V)                                                  | 29       |
| $I_{MP}$ (A)                                                  | 7.35     |
| Temperature Coefficient of $V_{OC}$ ( $\%/^{\circ}\text{C}$ ) | -0.36099 |
| Temperature Coefficient of $I_{SC}$ ( $\%/^{\circ}\text{C}$ ) | 0.102    |
| Number of Parallel Strings                                    | 17       |
| Number of Series Connected Modules Per String                 | 14       |

**Table 2.** PV panel Parameters.



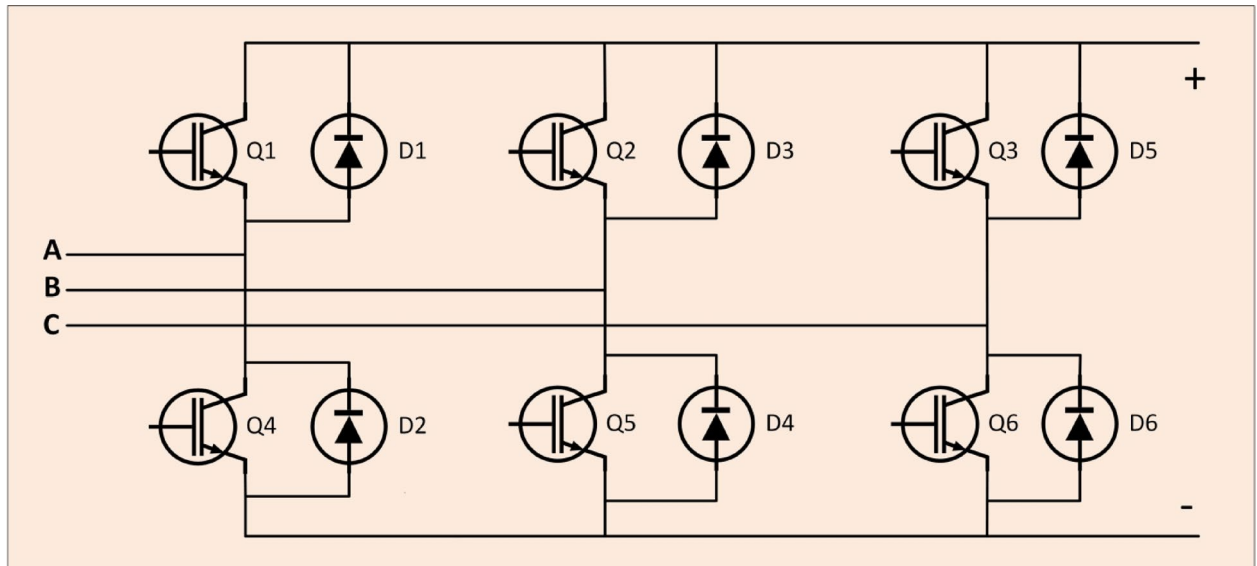
**Fig. 6.** Boost Converter.

#### Boost converter

The main idea of a boost converter is to step up the output voltage to a higher voltage level. The general overview of the boost converter is demonstrated in Fig. 6. The parameters calculation of the boost converter is given in Eqs. (2)–(5). The parameters of the boost converter are given in Table 3.



| Parameters | Value        |
|------------|--------------|
| $C_{in}$   | $100 \mu F$  |
| $C_0$      | $1000 \mu F$ |
| $L$        | $1.623 mH$   |
| $f_s$      | $5 kHz$      |

**Table 3.** Boost converter Parameters.**Fig. 7.** Inverter Topology.

| Parameters              | Value                  |
|-------------------------|------------------------|
| $R_{sn}$                | $100 k\Omega$          |
| Power Electronic Device | <i>IGBT with diode</i> |
| $R_{on}$                | $0.001 \Omega$         |
| Switching Frequency     | $5 kHz$                |

**Table 4.** Inverter Parameters.

$$D = 1 - \frac{V_{pv}}{V_o} \quad (2)$$

$$L = \frac{V_{pv} \times D}{f_s \times \Delta I_L} \quad (3)$$

$$C_0 = \frac{I_0 \times D}{f_s \times \Delta V_0} \quad (4)$$

$$R = \frac{V_0^2}{P} \quad (5)$$

In Table 3, boost converter design parameters are presented.

#### Inverter

Inverters play a critical role in PV energy systems by converting the DC output of solar panels into AC, which is necessary for compatibility with the conventional electrical grid. This DC–AC conversion is typically facilitated through high-frequency switching elements such as IGBTs and diodes. The inverter topology employed in this study is illustrated in Fig. 7, and its principal electrical specifications are detailed in Table 4.



### Inductive filter

In grid-connected PV systems, inductive filters are essential for suppressing high-frequency harmonics produced by inverter switching operations. These harmonics, if left untreated, can deteriorate power quality and lead to noncompliance with grid standards. The inductive filter also contributes to smoothing the output waveform, thereby ensuring more stable voltage and current delivery to the grid. In this simulation, an L-branch type inductive filter is employed. Inductance (L) of the inverter is 0.0027578 H<sup>65</sup>.

### Calculations of the performance metrics

The THD and efficiency calculations of the algorithms used in this study are formulated in Eqs. 6–8 respectively.

$$\text{THD} = \frac{I_H}{I_F} \quad (6)$$

$$I_H = \sqrt{I_2^2 + I_3^2 + \dots + I_n^2} \quad (7)$$

$$\text{Efficiency} = \left[ \frac{1}{T_s} \times \int_{t-T_s}^t 1 - \left( \frac{|\text{Ideal Power} - \text{Tracked Power}|}{\text{Ideal Power}} \right) \right] \times 100\% \quad (8)$$

Other performance metrics of relative power loss and relative power gain is formulated in Eq. 9<sup>66</sup>, and Eq. 10<sup>67</sup>, respectively.

$$\text{Relative Power Loss} = \frac{P_{pv, ideal} - P_{pv, real}}{P_{pv, ideal}} \times 100\% \quad (9)$$

$$\text{Relative Power Gain} = \frac{P_{pv, proposed} - P_{pv, base}}{P_{pv, base}} \times 100\% \quad (10)$$

### MPPT algorithms

The output power of PV panels depends on both solar irradiation and temperature, which continuously fluctuates over time. Without an MPPT algorithm, the boost converter may fail to maintain the correct duty cycle, leading to inefficient power conversion and unstable output. MPPT algorithms dynamically adjust the converter's duty cycle in real time to ensure operation at the MPP<sup>68</sup>.

#### Conventional algorithms

Conventional algorithms are the first algorithms for MPPT. They consist of simple mathematical equations and operations for the decision process. Therefore, the decision-making time of the algorithms is short. Although they have a simple structure, their sensitivity for deciding the true MPP is insufficient. Hence, the output power of the system oscillates around MPP, and the overall efficiency is low. Moreover, they cannot adapt to varying environmental conditions. Changing temperature and irradiance values make the decision process of algorithms difficult.

**P&O algorithm** The P&O algorithm is one of the commonly used MPPT algorithms in PV systems. This algorithm mainly depends on observing the output power of the system by perturbing the panel voltage and current. The output power is calculated over a time period, called the sample time. At the beginning of the algorithm, a small perturbation power value is calculated to provide a comparison with the next value. In a sample time, if the difference between the present and previous value ( $\Delta P$ ) is zero, the algorithm decides a non-changing perturbation. That means, the algorithm does not change the duty cycle ( $D$ ). If the algorithm detects a change between the present and previous power values, the voltage perturbation is observed. When the change of the perturbed voltage ( $\Delta V$ ) is negative, the algorithm increases the duty cycle. Conversely, if the change of the perturbed voltage is positive, the algorithm decreases the duty cycle<sup>69</sup>. The P&O algorithm is denoted in Fig. 8. The operating step size of the P&O algorithm is determined as 1e-05 s.

**INC algorithm** INC is a widely used and practical MPPT algorithm, especially suitable for rapidly changing environmental conditions. The algorithm begins by sensing the voltage and current of the PV panel. It then computes the instantaneous changes in voltage ( $\Delta V$ ) and current ( $\Delta I$ ) by comparing the current values with the previous ones. Using these, the algorithm calculates the instantaneous conductance ( $I/V$ ) and its derivative ( $\Delta I / \Delta V$ ). If there is no change in voltage ( $\Delta V = 0$ ), the algorithm evaluates  $\Delta I$  to determine the direction of movement. A positive  $\Delta I$  indicates that the operating point is to the left of the MPP, prompting the algorithm to decrease the voltage (i.e., reduce the duty cycle). Conversely, a negative  $\Delta I$  implies the operating point is to the right of the MPP, so the voltage is increased. If the derivative of the conductance ( $\Delta I / \Delta V$ ) equals the negative of the instantaneous conductance ( $-I/V$ ), the MPP is reached, and the duty cycle remains unchanged. If  $\Delta I / \Delta V > -I/V$ , the algorithm increases the voltage; otherwise, it decreases to<sup>70</sup>. The INC algorithm is illustrated in Fig. 9. Working step size of INC algorithm is determined as 1e-05 s.

#### Meta-heuristic algorithms

Meta-heuristic methods are inspired by the natural behaviour of humans, animals, and plants. These algorithms are more complex than conventional algorithms. They include more sensitive mathematical operations. However, they require specifically designed parameters for each system although these algorithms can have better power efficiency than conventional algorithms<sup>71</sup>.

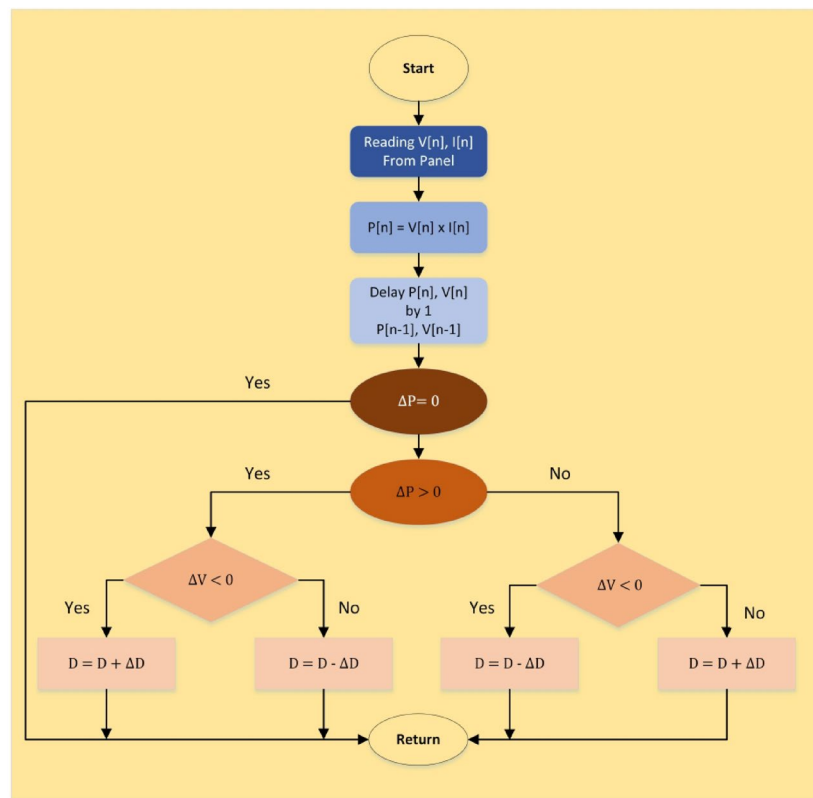


Fig. 8. P&amp;O Algorithm.

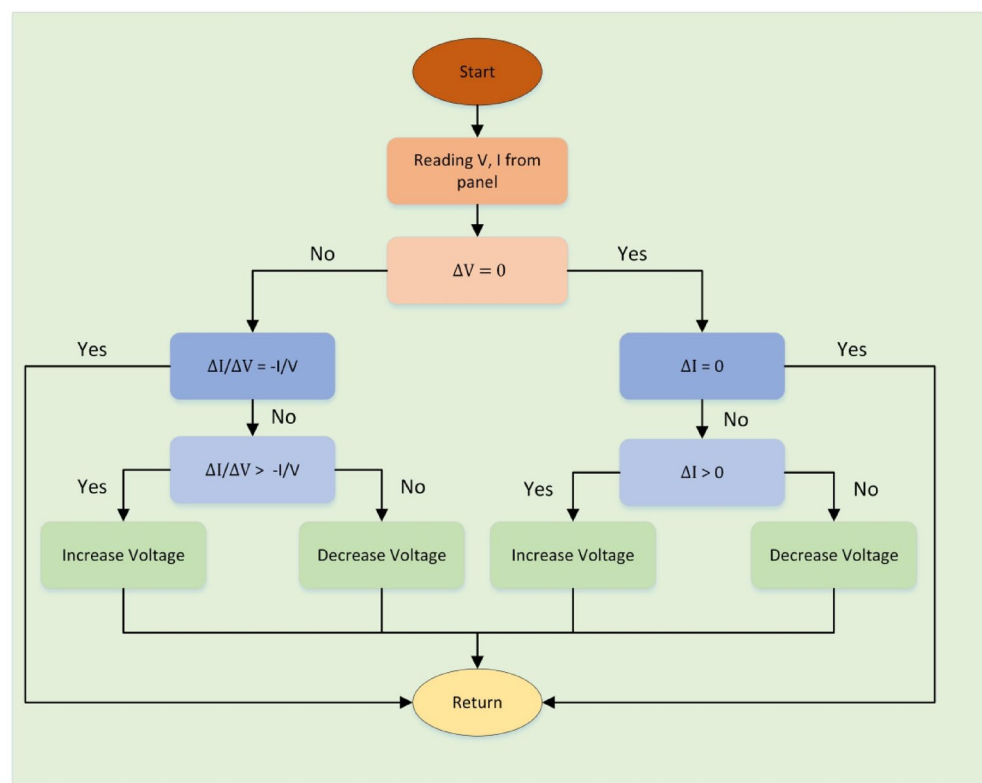
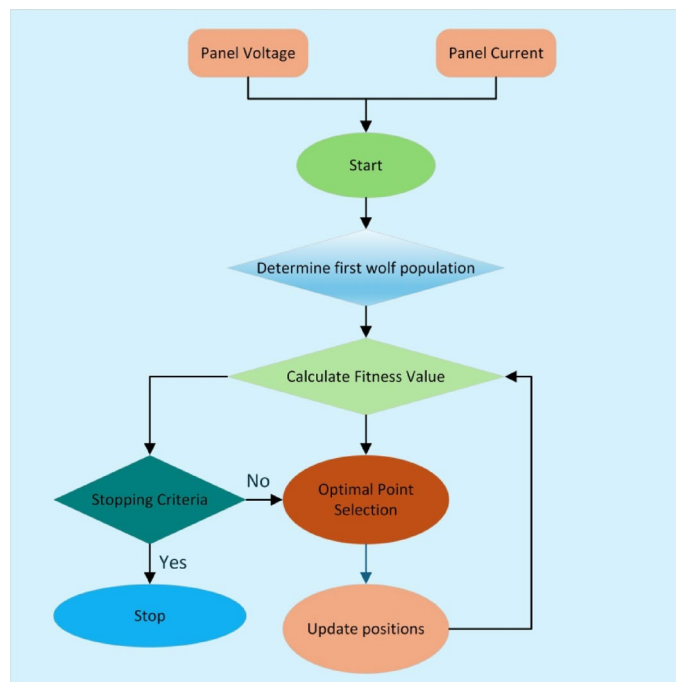


Fig. 9. INC Algorithm.



**Fig. 10.** GWO Algorithm.

| Parameters          | Value                                  |
|---------------------|----------------------------------------|
| Iteration           | 10                                     |
| Number of particles | 5 (four wolves and one for validation) |
| Step Size           | 1e-05 s                                |

**Table 5.** GWO algorithm Parameters.

**GWO algorithm** The GWO algorithm is inspired by the natural hunting behaviour of grey wolves. There are four wolves, each with a task in the hierarchy. First wolves are called the alpha ( $\alpha$ ), second wolves are called the beta ( $\beta$ ), third wolves are called the delta ( $\delta$ ), fourth wolves are called the omega ( $\omega$ ). The algorithm starts by determining the first two possible solutions for the alpha and beta wolves. Delta and omega wolves are responsible for providing solutions for alpha and beta. This algorithm has some stages, such as encircling prey, hunting, and attacking prey. Firstly, wolves try to encircle the global optimum power point (prey). Secondly, wolves start to move prey by learning from each other<sup>72</sup>. Finally, the fitness function reviews all results, adjusts the system to the best possible state, and prepares the following positions of the wolves. The fitness function finds the best MPP as shown in Eq. 11.

$$P_{pv} = V_{pv} \times I_{pv} \quad (11)$$

The flowchart of the GWO algorithm is illustrated in Fig. 10, and the parameters used in the algorithm are given in Table 5.

**FL algorithm** FL is one of the most widely used meta-heuristic MPPT techniques due to its simplicity and robustness under varying environmental conditions. In this method, the numerical inputs—typically voltage and current—are first transformed into linguistic variables through a process known as fuzzification, using pre-defined membership functions<sup>73</sup>. These functions map crisp input values (i.e., precise numerical data) to fuzzy sets with degrees of membership ranging from 0 to 1. Each input parameter has a membership function that defines terms such as “negative big,” “negative medium,” “zero,” “positive small,” and so on. The inference engine then evaluates the fuzzy rules and determines the appropriate fuzzy output based on a rule base. Finally, through the defuzzification process, the fuzzy output is converted back into a crisp control value, which is used to adjust the duty cycle of the converter to reach the GMPP<sup>74</sup>. The FL algorithm derives its decision-making ability from membership functions. The number of these functions and their correct parameterization are crucial. Incorrectly parameterized membership functions lead to a decrease in the algorithm’s efficiency<sup>75</sup>. Figure 11 describes the structure of the FL algorithm, and Table 6 gives the parameters used in the FL algorithm.

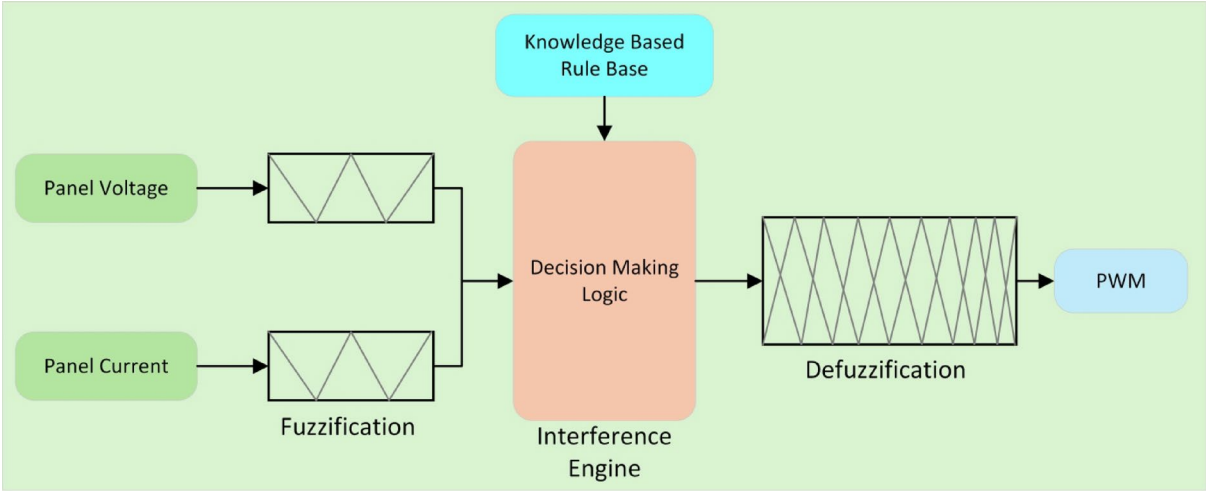


Fig. 11. FL Algorithm.

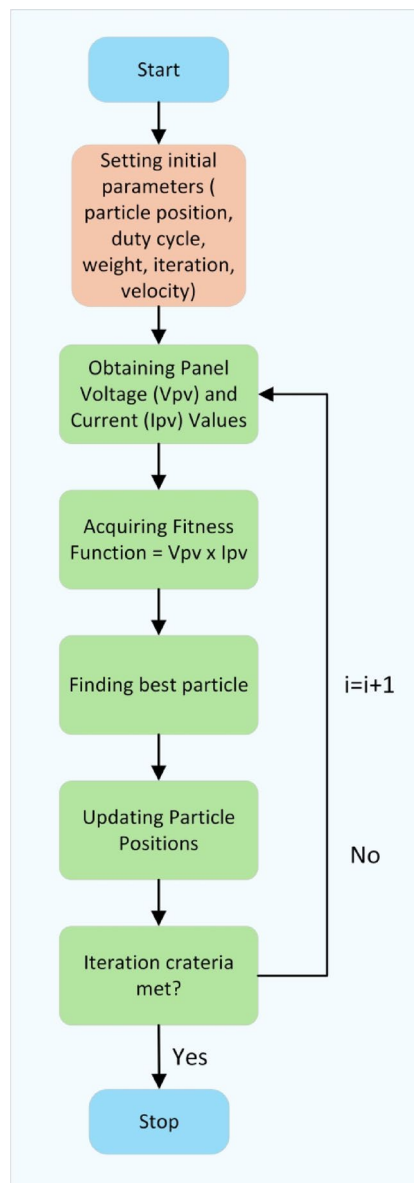
| Parameters                                           | Value   |
|------------------------------------------------------|---------|
| Number of the current input membership functions     | 5       |
| Number of the voltage input membership functions     | 7       |
| Number of the duty cycle output membership functions | 7       |
| Type of the membership functions                     | Trimf   |
| Step Size                                            | 1e-05 s |

Table 6. FL algorithm Parameters.

**PSO algorithm** The PSO algorithm is inspired by the regular behaviour of bird flocks. Birds flock together and intelligently position themselves and behave in a way that solves their own problems within the flock. Additionally, birds within flocks can further optimize their own positions by taking inspiration from each other. Inspired by these behaviors, the PSO algorithm aims to improve the position of each particle by taking into account each other's positions, which is called swarm intelligence<sup>76</sup>. At the beginning, as in every algorithm, the initial positions of the particles are determined, and these positions are candidates for the solution. In MPPT algorithms, these particles generally represent the duty cycle value. These particles optimize their positions both within themselves and within the swarm<sup>77</sup>. Additionally, each particle has its own speed to reach the optimum result. The Weight parameter determines how much of the particles' previous speeds are preserved. The Iteration coefficient determines how often this optimization process will be repeated at each sample time<sup>78</sup>. By determining these parameters, the particles change their positions by taking into account the increase and decrease of the output power and the positions of each other. However, these specified parameters need to be explicitly set for each system. In the conventional PSO algorithm, it is of great importance to determine these parameters appropriately for each system<sup>79</sup>, which is one of the most significant drawbacks of conventional PSO algorithms, as systems with variable output power, such as solar panels, require algorithms that can adapt to changing conditions. To address this issue, algorithms that optimize parameters already appear in the literature<sup>80</sup>. The flow diagram of the PSO algorithm is given in Fig. 12, and the parameters used in the study are indicated in Table 7.

*Learning based algorithms*

Learning-based approaches are primarily motivated by the capability to predict the MPP under varying climatic conditions, such as changes in solar irradiation. They are inspired by human neurons, which are used as layers in the systems. Before the process, data from panels and the environment are used to train the operation. After that, layers apply mathematical operations for elimination and estimation processes<sup>81</sup>. In this study, the dataset used for training and evaluating learning-based MPPT algorithms is generated from a PV energy conversion system consisting of a solar PV panel and a DC-DC boost converter. The input features of the dataset are the solar irradiance and cell temperature, which represent the environmental operating conditions of the PV system. The output variable is the optimized PV voltage corresponding to the maximum power point, obtained by controlling the duty cycle of the boost converter. By operating the system under a wide range of irradiance and temperature scenarios, a comprehensive dataset is constructed to capture the nonlinear mapping between environmental inputs and the optimal PV operating voltage. The control part diagram used in the study is given in Fig. 13. Also,  $K_p$  and  $K_i$  presented in Table 8.



**Fig. 12.** Flow diagram of the PSO algorithm.

| Parameters                    | Value   |
|-------------------------------|---------|
| Number of particles           | 8       |
| Iteration                     | 10      |
| Initial position of particles | 0.8     |
| Weight                        | 0.35    |
| Step Size                     | 1e-05 s |

**Table 7.** PSO algorithm Parameters.

**ANN algorithm** The central concept behind ANN algorithms is inspired by the functioning of the human brain. ANN consists of interconnected layers of artificial neurons that process and transmit information, mimicking biological neural structures. These networks typically include three types of layers: input, hidden, and output. The input layer receives data from external sources and passes it to the hidden layers, where most computations and pattern recognition occur. The output layer delivers the final result based on the processing in the hidden layers. During operation, each input is multiplied by a corresponding weight, and the weighted sum is then passed through an activation function in the hidden neurons, which determines whether a neuron becomes

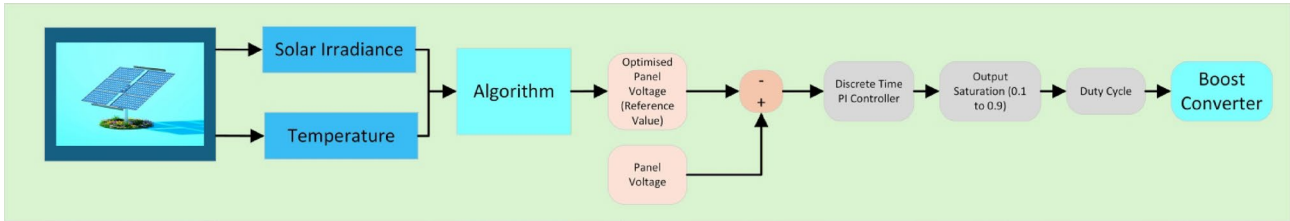


Fig. 13. Control Part Diagram.

| Parameters | Value |
|------------|-------|
| $K_p$      | 0.01  |
| $K_i$      | 0.2   |

Table 8. PI controller Parameters.

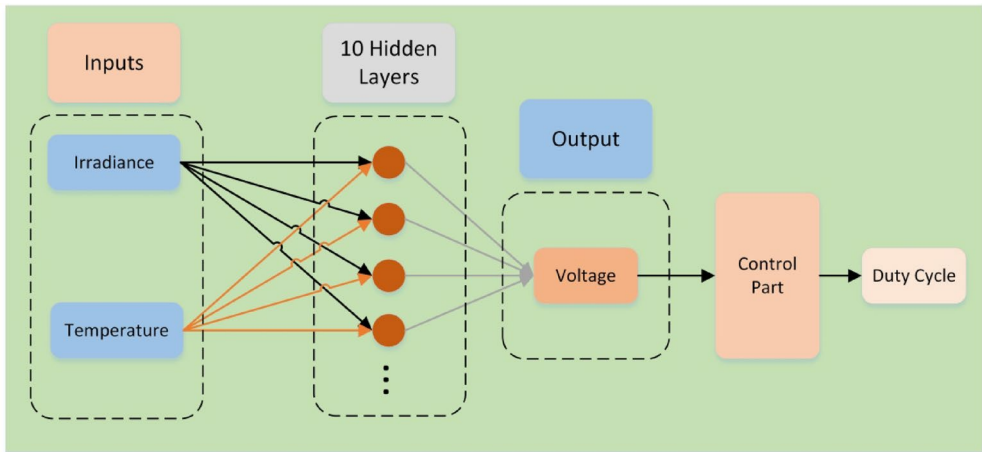


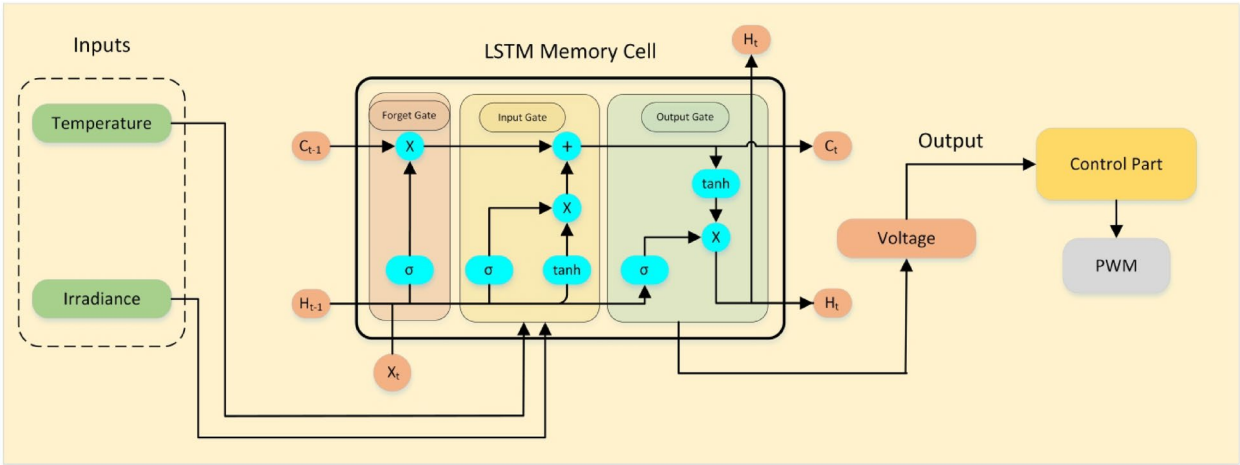
Fig. 14. ANN Algorithm.

“activated” based on its input. In feed-forward neural networks, the information flows in only one direction—from input to output—without feedback loops<sup>82</sup>. ANNs are well-suited for processing large datasets, as they can make rapid decisions by learning complex relationships between inputs and outputs. However, without adequate training data, the network may produce incorrect predictions or overgeneralizations. Therefore, ANN models require extensive and diverse datasets to ensure accurate and reliable performance. In this study, the inputs to the ANN include irradiance, temperature, panel current, and panel voltage, while the output is the power. The control logic then adjusts the system parameters accordingly. Figure 14 presents the structure of the ANN and Table 9 provides the parameters.

**LSTM algorithm** LSTM is one of the most widely used types of RNNs. LSTM networks are specifically designed to overcome the limitations of traditional RNNs, such as memory overflow and vanishing gradients. An LSTM network typically consists of four layers: a sequence input layer, an LSTM layer, a fully connected layer, and a regression layer<sup>83</sup>. Within the LSTM layer, each memory cell includes three gates: the forget gate, the input gate, and the output gate. These gates regulate the flow of information through the network using sigmoid ( $\sigma$ ) and tanh activation functions. The primary innovation of LSTM over traditional RNNs lies in the forget gate, which selectively removes irrelevant past data from memory. This mechanism helps reduce the system’s memory requirements and prevents the accumulation of unnecessary historical data, which could otherwise lead to inaccurate predictions<sup>84</sup>. Furthermore, LSTM-based MPP forecasting can improve system efficiency by minimising memory usage and reducing dependency on irrelevant past data. This approach directly addresses one of the significant disadvantages of RNNs—storing excessive outdated information<sup>85</sup>. However, although the LSTM algorithm deletes some past data from memory, it still remains insufficient for long-term prediction issues such as power monitoring. Moreover, due to their deep sequential structures, LSTM-based networks are computationally intensive and may pose challenges for real-time deployment in resource-constrained environments. Their “black-box” nature also limits interpretability and exposes them to risks of overfitting or underfitting de-

| Parameters              | Value                                |
|-------------------------|--------------------------------------|
| Number of hidden layers | 10                                   |
| Training data           | 75%                                  |
| Validation data         | 15%                                  |
| Test data               | 10%                                  |
| Learning rate           | 0.0001                               |
| Training Environment    | MATLAB – Neural Network Fitting Tool |

**Table 9.** ANN algorithm training Parameters.



**Fig. 15.** LSTM Algorithm.

| Parameters                | Value                                              |
|---------------------------|----------------------------------------------------|
| Number of hidden layers   | 10                                                 |
| Training data             | 90%                                                |
| Test data                 | 10%                                                |
| Learning schedule         | Custom warmup                                      |
| Initial learning rate     | 0.0001                                             |
| Learning rate drop period | 10                                                 |
| Learning rate drop factor | 0.01                                               |
| Training method           | MATLAB Adam (Adaptive moment estimation) optimizer |

**Table 10.** LSTM algorithm training Parameters.

pending on training conditions. In particular, it is stated that the LSTM algorithm has problems in complex and long-term power tracking with multiple timesteps. New algorithms have been developed to solve these problems encountered in the LSTM algorithm<sup>86</sup>. In this study, the inputs to the LSTM network are defined as temperature and irradiance, while the output is the panel voltage. A control unit then uses the LSTM’s output to determine the optimal duty cycle. The LSTM algorithm is illustrated in Fig. 15, and the parameters used in the algorithm are given in Table 10.

**BiLSTM algorithm** The BiLSTM algorithm is a bidirectional application of the LSTM algorithm. By applying the LSTM algorithm in both forward and backward directions, more optimized results are achieved, and faster learning is possible during long-term data tracking<sup>87</sup>. While the forward LSTM algorithm provides benefits for predicting future situations, the backward LSTM algorithm maximizes the efficiency of the decision to be made at the output by taking inspiration from past situations<sup>88</sup>. The BiLSTM algorithm also improves system performance in unexpected situations by increasing the system’s learning content. A high degree of learning data is particularly advantageous in cases of PSC in solar systems<sup>89</sup>. However, the weight parameter must be updated at the same rate in the predictions, because continually using the same weight parameter makes the system vulnerable to extreme situations<sup>90</sup>. Figure 16 illustrates the operation performed in a cell of the BiLSTM algorithm, and Table 11 illustrates the parameters of the BiLSTM algorithm used in this study.



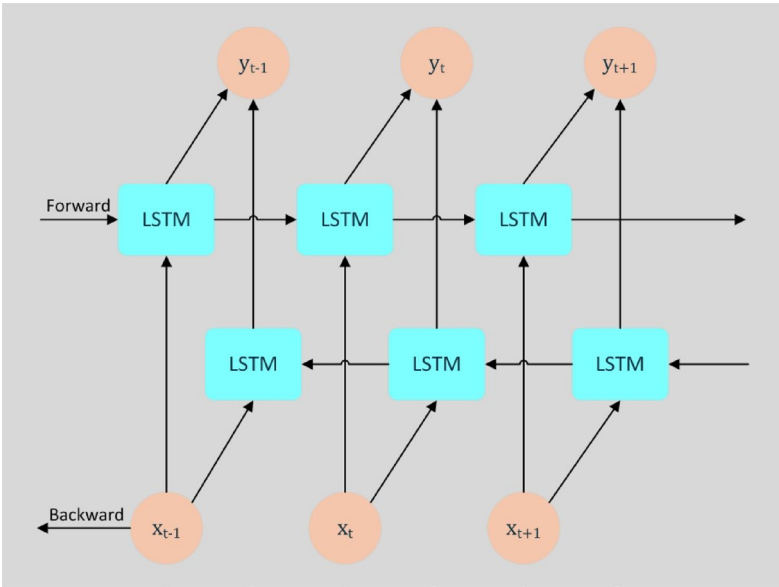


Fig. 16. BiLSTM Algorithm.

| Parameters                | Value                                              |
|---------------------------|----------------------------------------------------|
| Number of hidden layers   | 10                                                 |
| Training data             | 90%                                                |
| Test data                 | 10%                                                |
| Number of epochs          | 250                                                |
| Learning schedule         | Custom warmup                                      |
| Initial learning rate     | 0.005                                              |
| Learning rate drop period | 10                                                 |
| Learning rate drop factor | 0.02                                               |
| Training method           | MATLAB Adam (Adaptive moment estimation) optimizer |

Table 11. BiLSTM algorithm training Parameters.

**ANFIS algorithm** The ANFIS algorithm is a hybrid approach that integrates ANN with FL, thereby leveraging the complementary advantages of both methodologies. By integrating the pre-trained nature and deep decision-making capability of ANN with the rule-based decision-making strength of FL, ANFIS emerges as a powerful and effective algorithm<sup>91</sup>. The ANFIS algorithm begins with optimizing parameters using an ANN. Therefore, the ANFIS algorithm includes input, hidden, and output layers similar to the ANN algorithm or MPPT in solar panel systems, irradiation, temperature, voltage, and current can be used as input layers. The algorithm’s output can be voltage, current, or duty cycle<sup>92</sup>. After the ANN operation, which is evaluated through membership functions. Membership functions can be of various types; for example, a trimf is used in this study. The ANFIS algorithm, with its detailed decision-making and ability to work with large datasets, is used to solve many problems<sup>93</sup>. Additionally, the FL algorithm has a high ability to perform well in uncertain issues. This feature, combined with ANN’s training ability, creates a perfect synergy<sup>94</sup>. The ANFIS algorithm is illustrated in Fig. 17, and the parameters used in this study are illustrated in Table 12.

### Results and discussion

Figure 18 demonstrates a MATLAB simulation model of a grid-connected PV system including MPPT algorithms.

All algorithms are simulated for 4 s in MATLAB/Simulink, as shown in Fig. 18, using a sample time of 1e-05 s. The overall performance, efficiency, THD values, and duty-cycle plots of the algorithms are presented in Fig. 19 for the sinusoidal solar radiation test, and in Fig. 20 for the PSC. Among these parameters, the performance metric represents the comparison between the ideal output power of the system and the power output obtained in the simulation. The efficiency value indicates how accurately each algorithm is able to track the MPP; the THD values represent the total harmonic distortion occurring in the system’s inverter; and the duty-cycle values show the duty cycle at which the IGBT in the system is driven by the microcontroller during MPP tracking.

Additionally, the efficiency, THD, relative power loss and relative power gain values presented in Tables 13 and 14 are calculated using Eqs. (6)–(10). The relative power loss values indicate how much power the system

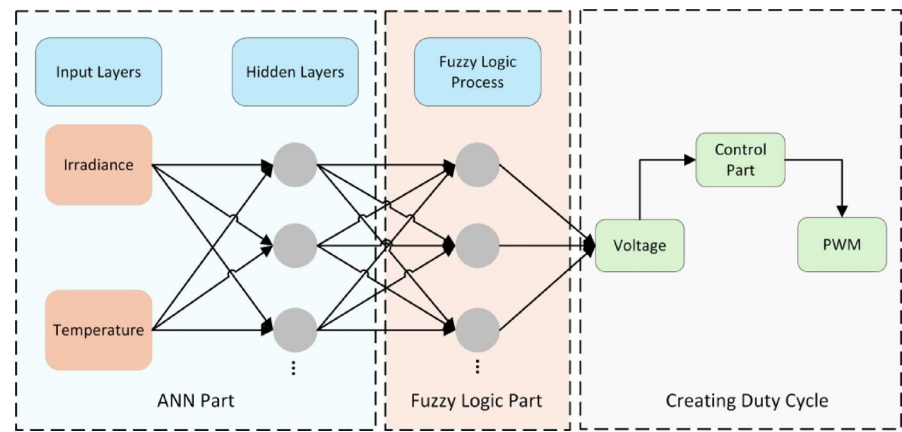


Fig. 17. ANFIS Algorithm.

| Parameters                     | Value                             |
|--------------------------------|-----------------------------------|
| Number of membership functions | 10                                |
| Number of epochs               | 100                               |
| Type of membership functions   | Trimf                             |
| Error tolerance                | 0                                 |
| Training method                | Grid partition                    |
| Training environment           | MATLAB, Neuro-Fuzzy Designer Tool |

Table 12. ANFIS algorithm training Parameters.

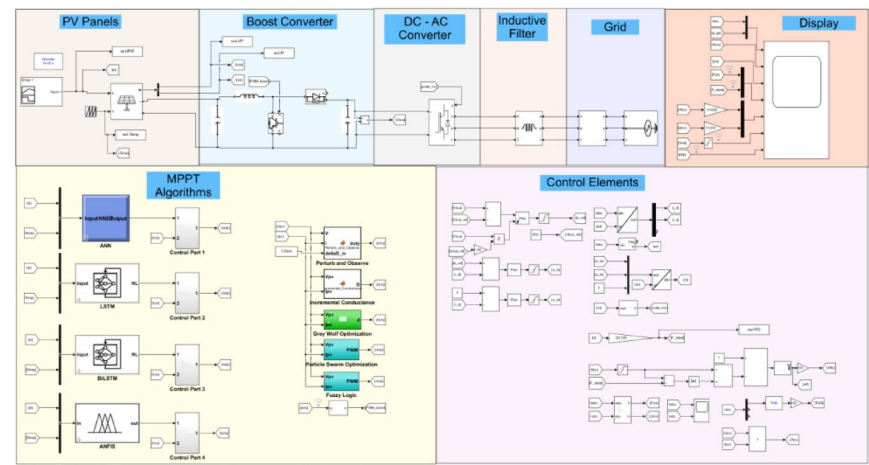
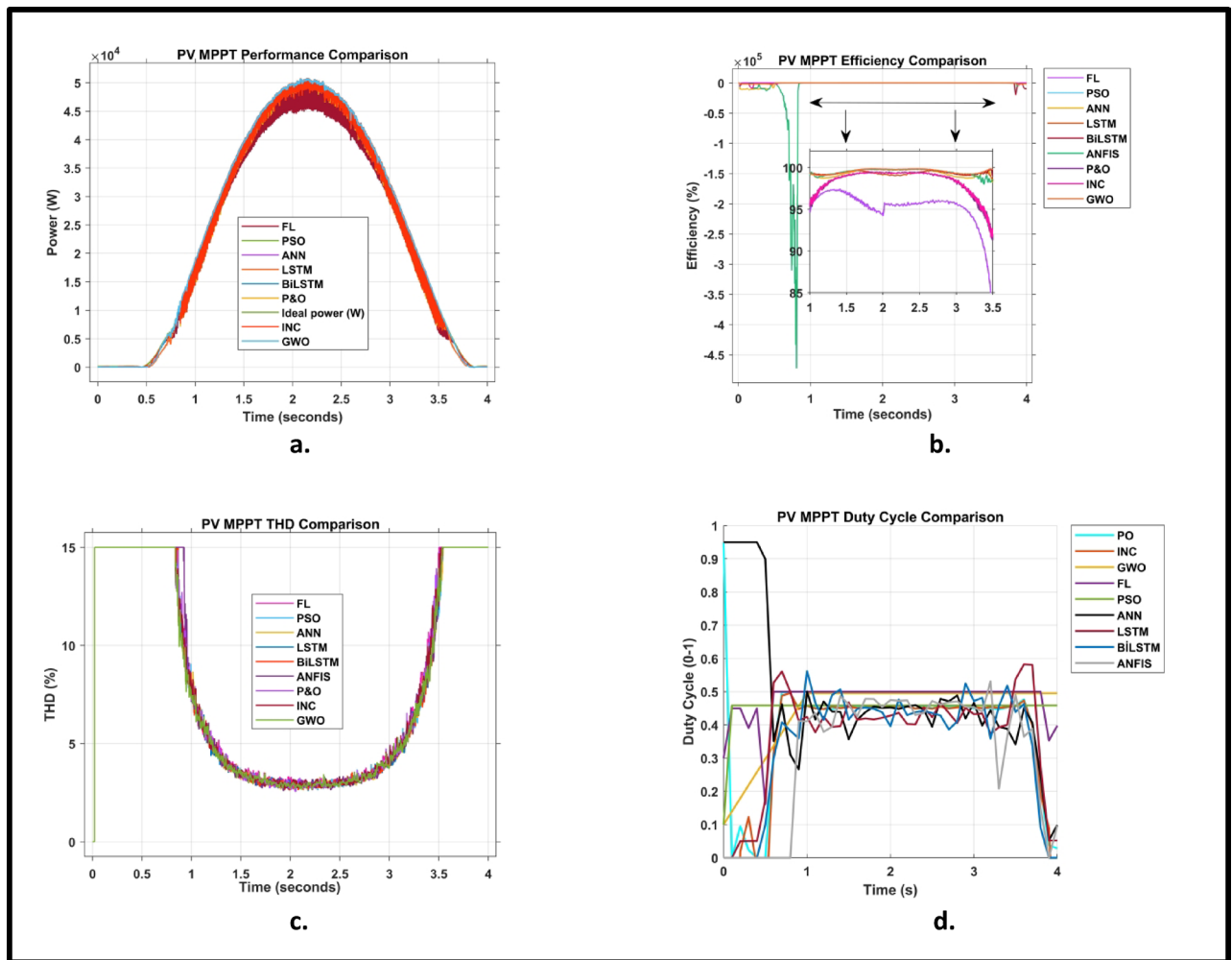


Fig. 18. Simulation Model for Grid-Connected PV System.

loses during operation, whereas the relative power gain value shows the improvement achieved by each algorithm compared to the one with the lowest efficiency, which is designated as the base algorithm in the table. The computational complexity value indicates how much processing power each algorithm requires, expressed in terms of low, medium, or high.

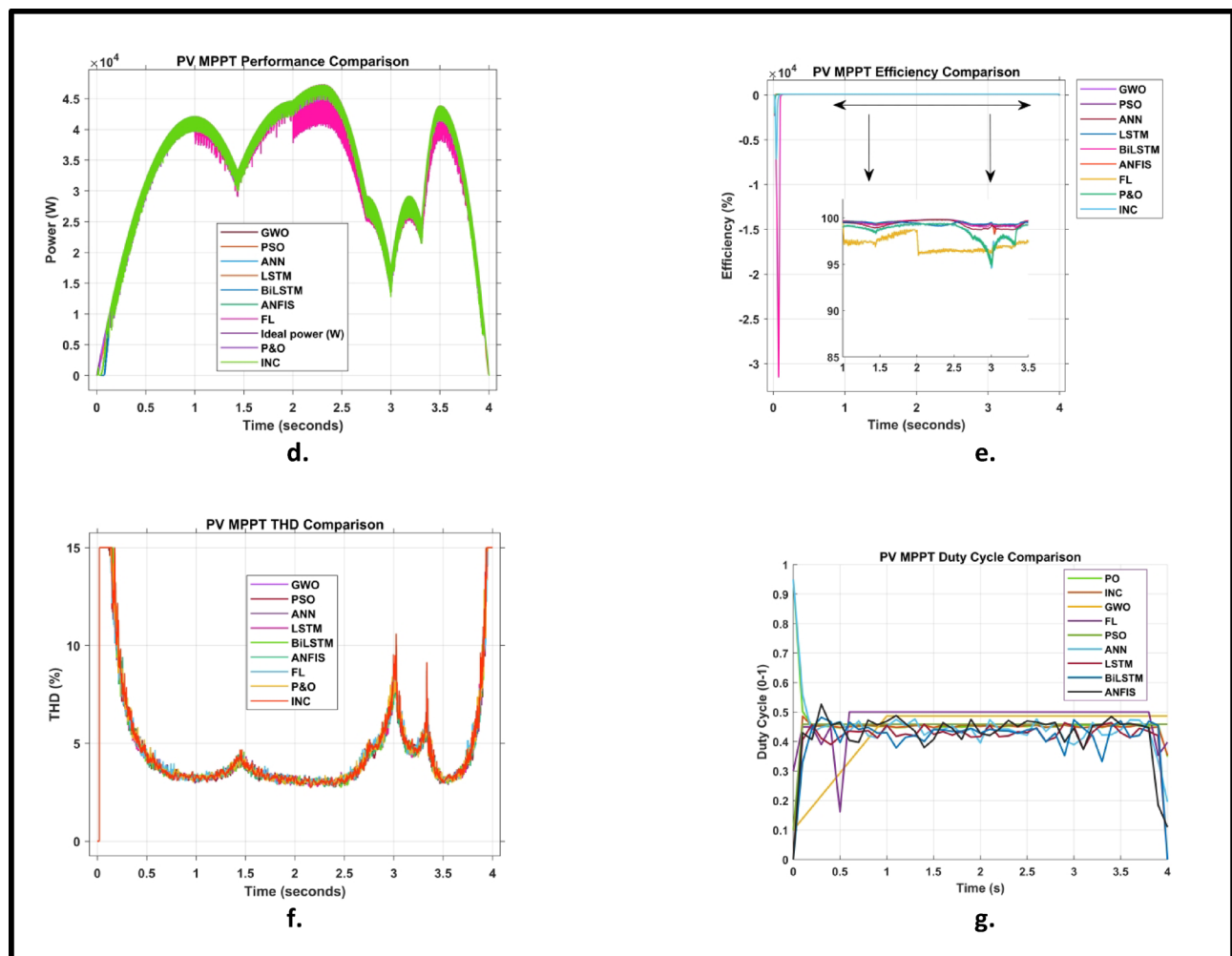
According to the results summarized in Table 13, under the sinusoidal irradiation test condition, the conventional P&O and INC algorithms exhibit efficiencies of 98.28% and 98.30%, respectively. In contrast, the metaheuristic and intelligence-based approaches demonstrate superior performance, with GWO and PSO achieving the highest efficiency of 99.53%, followed closely by BiLSTM (99.51%), ANFIS (99.43%), ANN (99.36%), and LSTM (99.31%). The FL-based method yields the lowest efficiency at 95.23%. In terms of power quality, the THD values remain within a narrow range for all algorithms. The lowest THD levels are observed for GWO, ANN, LSTM, and ANFIS at 4.21%, while the P&O, INC, and FL methods exhibit slightly higher THD values of 4.31%. These results indicate that advanced optimization and learning-based MPPT techniques can



**Fig. 19.** PV MPPT Sinus Irradiance Test Simulation Results (a. Performance Comparison, b. Efficiency Comparison, c. THD Comparison, d. Duty Cycle Comparison).

marginally improve harmonic performance. Regarding steady-state oscillations, conventional methods such as P&O and INC suffer from relatively large oscillation amplitudes of approximately 4 kW, whereas FL exhibits the highest oscillation at 4.5 kW. In contrast, intelligent and metaheuristic approaches significantly reduce oscillations, with ANN, LSTM, BiLSTM, and ANFIS limiting power fluctuations to around 240 W, and GWO and PSO achieving oscillation levels of 260 W and 250 W, respectively. The relative power loss analysis further confirms the superiority of advanced techniques. The FL method experiences the highest power loss at 4.77%, while P&O and INC incur losses of 1.72% and 1.70%, respectively. In comparison, GWO and PSO achieve the minimum relative power loss of 0.47%, followed by BiLSTM (0.49%), ANFIS (0.57%), ANN (0.64%), and LSTM (0.69%). Correspondingly, the relative power gain is maximized for GWO and PSO at 4.51%, with BiLSTM (4.49%) and ANFIS (4.41%) also providing substantial gains over the baseline FL approach. Finally, from a computational complexity perspective, P&O and INC are classified as low-complexity methods, making them suitable for low-cost implementations. GWO and PSO exhibit medium computational complexity, offering a favorable trade-off between performance and implementation effort. In contrast, FL, ANN, LSTM, BiLSTM, and ANFIS are categorized as high-complexity algorithms, which may impose higher computational and hardware requirements despite their enhanced tracking performance.

According to the results reported in Table 14, under PSC, all MPPT algorithms exhibit a general improvement in efficiency compared to uniform irradiation scenarios; however, notable performance disparities persist among conventional, metaheuristic, and learning-based methods. The conventional P&O and INC algorithms achieve identical efficiencies of 98.64%, whereas the FL-based approach demonstrates a comparatively lower efficiency of 97.10%. In contrast, advanced techniques deliver superior performance, with ANFIS attaining the highest efficiency of 99.51%, followed closely by BiLSTM (99.44%), LSTM (99.42%), GWO and PSO (99.49%), and ANN (99.29%). From a power quality perspective, the THD values remain consistently low across all methods, indicating stable converter operation under PSC. The minimum THD values of 3.86% are achieved by GWO, PSO, and BiLSTM, while ANN and ANFIS present slightly higher yet comparable THD levels of 3.88%. Conventional P&O, INC, and FL methods exhibit marginally higher THD values, reaching up to 3.96%



**Fig. 20.** PV MPPT PSC Test Simulation Results (d. Performance Comparison, e. Efficiency Comparison, f. THD Comparison, g. Duty Cycle Comparison).

| Algorithm | Efficiency | THD   | Oscillation | Relative Power Loss | Relative Power Gain | Computational Complexity |
|-----------|------------|-------|-------------|---------------------|---------------------|--------------------------|
| P&O       | 98.28%     | 4.31% | 4 kW        | 1.72%               | 3.21%               | Low                      |
| INC       | 98.30%     | 4.31% | 4 kW        | 1.7%                | 3.22%               | Low                      |
| GWO       | 99.53%     | 4.21% | 260 W       | 0.47%               | 4.51%               | Medium                   |
| FL        | 95.23%     | 4.31% | 4.5 kW      | 4.77%               | Base                | High                     |
| PSO       | 99.53%     | 4.24% | 250 W       | 0.47%               | 4.51%               | Medium                   |
| ANN       | 99.36%     | 4.21% | 240 W       | 0.64%               | 4.33%               | High                     |
| LSTM      | 99.31%     | 4.21% | 240 W       | 0.69%               | 4.28%               | High                     |
| BiLSTM    | 99.51%     | 4.22% | 240 W       | 0.49%               | 4.49%               | High                     |
| ANFIS     | 99.43%     | 4.21% | 240 W       | 0.57%               | 4.41%               | High                     |

**Table 13.** Results of the MPPT algorithms in sinus irradiation Test.

in the case of FL, though still within acceptable limits. The steady-state oscillation analysis reveals significant differences in dynamic behavior. Conventional algorithms such as P&O and INC continue to suffer from large oscillation amplitudes of approximately 4 kW, with the FL method exhibiting the highest oscillation level of 4.5 kW. Conversely, metaheuristic and intelligent algorithms markedly suppress power oscillations, limiting steady-state fluctuations to 260 W for GWO, 250 W for PSO, and approximately 240 W for ANN, LSTM, BiLSTM, and ANFIS. This substantial reduction highlights the robustness of learning-based approaches in tracking the global maximum power point under PSC. The relative power loss assessment further corroborates these findings. The highest power loss is observed for the FL algorithm at 2.90%, whereas P&O and INC incur losses of 1.36%. In contrast, ANFIS achieves the minimum relative power loss of 0.49%, followed by GWO and PSO at

| Algorithm | Efficiency | THD   | Oscillation | Relative Power Loss | Relative Power Gain | Computational Complexity |
|-----------|------------|-------|-------------|---------------------|---------------------|--------------------------|
| P&O       | 98.64%     | 3.92% | 4 kW        | 1.36%               | 1.58%               | Low                      |
| INC       | 98.64%     | 3.94% | 4 kW        | 1.36%               | 1.58%               | Low                      |
| GWO       | 99.49%     | 3.86% | 260 W       | 0.51%               | 2.46%               | Medium                   |
| FL        | 97.10%     | 3.96% | 4.5 kW      | 2.90%               | Base                | High                     |
| PSO       | 99.49%     | 3.86% | 250 W       | 0.51%               | 2.46%               | Medium                   |
| ANN       | 99.29%     | 3.88% | 240 W       | 0.77%               | 2.25%               | High                     |
| LSTM      | 99.42%     | 3.87% | 240 W       | 0.58%               | 2.38%               | High                     |
| BiLSTM    | 99.44%     | 3.86% | 240 W       | 0.56%               | 2.40%               | High                     |
| ANFIS     | 99.51%     | 3.88% | 240 W       | 0.49%               | 2.48%               | High                     |

**Table 14.** Results of the MPPT algorithms in PSC.

| Scenarios                                                      | Operating Conditions                                        | Performance Priority                               | Suggested Algorithms             |
|----------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------|----------------------------------|
| S1: Steady environment                                         | Low irradiance variation, no PSC                            | Fast convergence, low complexity                   | P&O, and INC                     |
| S2: Low hardware power                                         | Microcontroller-based, limited CPU/RAM                      | Minimum memory and processing load                 | P&O, and INC                     |
| S3: Rapidly changing irradiance                                | If there are sudden radiation fluctuations                  | Dynamic tracking speed must be high                | PSO, and GWO                     |
| S4: Intense PSC                                                | Multiple GMPP status                                        | GMPP tracking success                              | LSTM, and BiLSTM                 |
| S5: Grid-connected and power quality-prioritized systems       | THD should be low, and emissions should be limited.         | Minimizing harmonic distortion                     | PSO, GWO+ parameter optimisation |
| S6: Moderately variable environments + training data available | Trainable artificial intelligence models can be implemented | Balanced performance, no need to feedback networks | ANN, and ANFIS                   |
| S7: Data-driven, high-processing advanced systems              | GPU/FPGA current, historical rich data                      | Highest predictive accuracy                        | LSTM, BiLSTM, and ANFIS          |

**Table 15.** MPPT algorithms comparison in different Scenarios.

0.51%, and BiLSTM and LSTM at 0.56% and 0.58%, respectively. These reduced loss values directly translate into enhanced energy harvesting performance under nonuniform irradiation conditions. Consistently, the relative power gain analysis indicates that advanced methods significantly outperform the baseline FL approach. ANFIS provides the highest relative power gain of 2.48%, followed by BiLSTM (2.40%), LSTM (2.38%), and GWO/ PSO (2.46%). ANN also demonstrates a notable improvement with a relative gain of 2.25%, while conventional P&O and INC offer limited gains of 1.58%. Finally, in terms of computational complexity, P&O and INC remain low-complexity algorithms suitable for simple and cost-sensitive applications. GWO and PSO are classified as medium-complexity methods, offering an effective compromise between performance and computational burden. In contrast, FL, ANN, LSTM, BiLSTM, and ANFIS exhibit high computational complexity, which may impose higher processing requirements but is justified by their superior tracking accuracy and robustness under PSC.

Table 15 summarizes the operating scenarios considered for the MPPT algorithms, the corresponding key performance metrics evaluated under each scenario, and the MPPT techniques applicable to these operating conditions. Table 16 provides a detailed assessment of the computational complexity of the investigated algorithms, including hardware requirements, inference latency, memory usage, and real-time deployability. The P&O and the INC algorithms exhibit lower efficiency than other methods. In addition, they tend to produce higher levels of oscillation. This limitation arises because the mathematical functions underlying P&O and INC are not sufficiently precise in tracking the MPP relative to more advanced algorithms. However, their advantage lies in their simplicity, as they require minimal computational power and can be easily integrated into the system. The low computational complexity of the P&O and INC algorithms stems from the absence of a specific iteration parameter; therefore, they execute only once per sampling time. On the other hand, the FL algorithm exhibits high computational complexity due to its use of membership functions and rule-based inference mechanisms. Moreover, metaheuristic algorithms such as GWO and PSO demonstrate high efficiency in power tracking, owing to their superior decision-making capabilities. However, these algorithms require parameter tuning for each specific system and impose a higher computational load compared to conventional algorithms. The reason that the PSO and GWO algorithms have a medium computational load is that they perform decision-making over multiple iterations for each sampling time. Learning-based algorithms, including ANN, LSTM, BiLSTM, and ANFIS, have demonstrated high efficiency, which is primarily because these algorithms are trained on data, enabling them to anticipate operating conditions such as PSC. Additionally, their hidden layers provide enhanced decision-making capabilities. Compared to feed-forward networks such as ANN, feedback networks, including LSTM and BiLSTM, stand out due to their ability to exploit temporal dependencies and leverage past information, further improving performance. Although ANFIS is not a deep learning or feedback-based network, it demonstrates high efficiency because the combination of ANN and FL endows the algorithm with strong decision-making capabilities. However, although learning-based algorithms are highly efficient and resilient to PSCs, they are inherently complex and require substantial computational resources, which is mainly

| Algorithm | Typical Hardware          | Inference Latency (μs) | Memory Usage             | Real-Time Deployability |
|-----------|---------------------------|------------------------|--------------------------|-------------------------|
| P&O       | Low-end MCU (8–16 bit)    | < 10                   | Very Low (KB-level)      | High                    |
| INC       | Low-end MCU (16–32 bit)   | < 15                   | Very Low (KB-level)      | High                    |
| PSO       | 32-bit MCU/DSP            | 40–150 *               | Medium                   | Medium                  |
| GWO       | 32-bit MCU/DSP            | 35–140 *               | Medium                   | Medium                  |
| FL        | 32-bit MCU/DSP            | 80–300**               | Medium                   | Medium                  |
| ANN       | DSP/ARM Cortex-M4/M7      | 40–150                 | High (tens of KB)        | Low-Medium              |
| LSTM      | ARM Cortex-A/Embedded GPU | 200–800                | High (hundreds of KB-MB) | Low                     |
| BiLSTM    | ARM Cortex-A/Embedded GPU | 400–1500               | High (MB-level)          | Low                     |
| ANFIS     | DSP/ARM Cortex-M7         | 100–350                | High (MB-level)          | Low                     |

**Table 16.** MPPT algorithms computational complexity Comparison. \*Latency depends on population size and iteration number. \*\*Latency increases with the number of fuzzy rules.

attributed to the number of network parameters and inference operations, resulting in increased memory usage and processing requirements for real-time implementation.

Conclusion

MPPT algorithms play a critical role in enhancing the efficiency of PV systems by continuously optimising power output. Different MPPT strategies offer varying levels of performance in terms of efficiency, computational complexity, and power quality. A systematic comparison is essential to identify the most suitable method for specific operational requirements and conditions. The current study conducted a comprehensive comparative evaluation of multiple MPPT algorithms within a grid-connected PV system. The research incorporated conventional algorithms (P&O and INC), meta-heuristic intelligent methods (FL, GWO, and PSO), and learning-based models (ANN, LSTM, BiLSTM, and ANFIS). The simulation environment is developed in MATLAB/Simulink, and performance is evaluated based on tracking efficiency, oscillation amplitude, computational complexity, relative power loss, relative power gain, and THD. The algorithms are subjected to two different tests: the sinusoidal solar radiation test and the PSC test. Results are obtained for both tests and subsequently compared. The results demonstrated that although conventional methods such as P&O and INC achieved satisfactory tracking efficiency in both the sinus irradiation test and the PSC test, they suffered from significant oscillations and limited adaptability to rapidly changing environmental conditions. Their simplistic control structures—while computationally efficient—render them less effective in scenarios with dynamic irradiance and temperature profiles.

The meta-heuristic optimisation-based methods revealed noticeable improvements. GWO and PSO offered lower oscillations and higher stability due to their adaptive nature and population-based optimisation strategy. However, these algorithms impose a higher computational load compared to conventional methods, and their parameters must be individually tuned for each system, complicating their integration. In comparison with PSO and GWO, the FL algorithm requires greater computational effort and exhibits lower efficiency than the other algorithms. The limitations of metaheuristic algorithms can be addressed by hybridizing them with other conventional metaheuristic methods. Among learning based methods, the ANN-based MPPT algorithm can provide effective solutions in power tracking despite being a feed-forward network. In contrast, networks incorporating feedback, such as LSTM and BiLSTM, can achieve maximum performance, particularly under conditions like PSC, due to their ability to learn from past events. Although ANFIS does not include a feedback network, it demonstrates high performance owing to the deep decision-making capabilities inherited from its constituent algorithms. However, a key limitation of these algorithms is their high computational demand. Future research could explore hybridizing deep learning PV system MPPT algorithms with other metaheuristic methods to reduce their computational load.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author (yunusyalman@aybu.edu.tr) on reasonable request.

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## Author contributions

Conceptualization, Y.Y., K.O., methodology, Y.Y., M. D.; software, M.D.; validation, M.D., K.O and Y.Y.; formal analysis, Y.Y.; investigation, M.D, K.O.; resources, M.D.; writing—original draft preparation, M.D.; writing—review and editing, Y.Y., K.O; visualization, M.D.; supervision, Y.Y., K.O. All authors have read and agreed to the published version of the manuscript.

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## Declarations

## Competing interests

The authors declare no competing interests.

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